

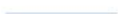
LLM Foundations and Prompt Design

Build grounded LLM experiences with
measurable quality and safeguards.



Learning outcomes

01  Explain tokens, context, sampling, and limitations.

02  Design structured prompts with testable outputs.

03  Evaluate representative cases.

The concept system

01

tokens

02

context windows

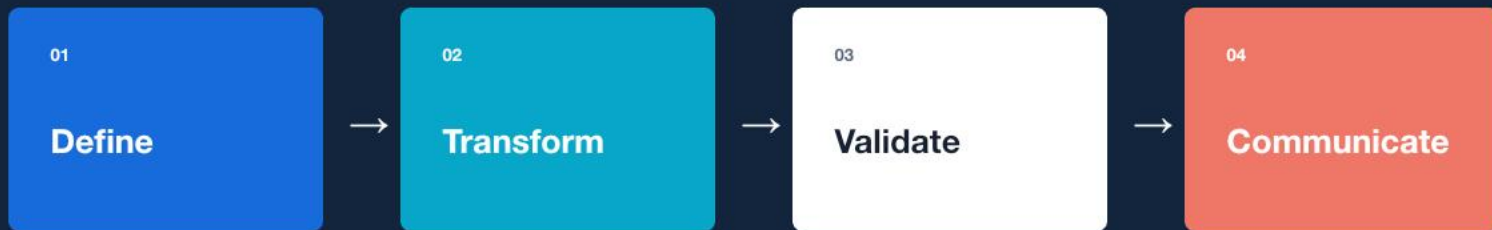
03

prompt contracts

04

evaluation

A repeatable reasoning loop



Create a structured data-quality assistant with an evaluation set.

Guided lab

LAB BRIEF

Create a structured data-quality assistant with an evaluation set.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A prompt specification, test cases, and failure analysis.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Grounded Knowledge Assistant

Retrieval 25%, grounding 25%, evaluation 20%, safeguards 20%, documentation 10%.

What should remain after the lesson?

CONCEPT

tokens + context windows

PRACTICE

Create a structured data-quality assistant with an evaluation set.

EVIDENCE

A prompt specification, test cases, and failure analysis.
